

Ubuntu Basic Commands

1. Basic System Information and Navigation

Understanding the system

hostname	# Display the name of the machine
who	# Show logged-in users
date	# Show current date and time
man uname	# Manual for uname command

Navigating the File System

pwd	# Show the current directory path
ls	# List files in the current directory
ls -l	# Detailed list view with permissions
ls -a	# Show hidden files
ls -t	# Sort files by modification time
ls -R	# Recursively list directory contents
ls -r	# Print in reverse alphabetical order
man ls	# Manual for ls command
cd test_dir/	# Move into a directory
cd ..	# Go back to the previous directory
mkdir new_dir	# Create a new directory
Mkdir -p new_dir	# Overwrite a directory if already existed
rmdir new_dir	# Remove an empty directory

2. File Management

Creating and Editing Files

touch file.txt	# Create an empty file
echo 'Hello Linux' > file.txt	# Create and write into a file
echo 'Hello world' >> file.txt	# Append text to a file

Copying, Moving, and Deleting Files

cp file.txt copy_file.txt	# Copy file to another file
mkdir dir	# Create a directory
mkdir -p dir	# Create parent directory if needed
cp file.txt dir/	# Copy file to a directory
mv file.txt renamed_file.txt	# Rename or move a file
rm renamed_file.txt	# Delete a file
rmdir dir	#Delete empty directory
rm -r dir/	# Delete a directory and its contents

Searching for Files

find / -name file.txt	# Search for file.txt in the entire system
find . -type f -name "*.txt"	# Find all .txt files in the current directory
locate file.txt	# Quickly find a file

3. User and Permission Management

Managing File Permissions

ls -l file.txt	# Check file permissions
chmod u+x file.txt	# Give execute permission to the user
chmod go+x file.txt	# Give execute permission to group C others
chmod -x file.txt	# Remove execute permission
chmod 766 file.txt	# Set specific permissions
sudo chown student file.txt	# Change file ownership sudo
chgrp group file.txt	# Change group ownership

4. Viewing and Processing Files

Displaying File Content

cat file.txt	# Display file contents
head -n file.txt	# Show the first n lines
tail -5 file.txt	# Show the last 5 lines

Sorting and Counting Content

sort file.txt	# Sort file contents
uniq file.txt	
wc file.txt	# Count lines, words, and characters
wc -w file.txt	# Count words only

Searching Within Files (grep Commands)

grep 'Linux' file.txt	# Search for 'Linux' in file.txt
grep -i 'linux' file.txt	# Case-insensitive search
grep -n 'Linux' file.txt	# Show line numbers for matches
grep -c 'Linux' file.txt	# Count occurrences of 'Linux'
grep -q 'Linux' file.txt	# Only outputs 0 if match found, 1 if no match
grep -w 'Linux' file.txt	# Searches only for the whole word

5. Advanced File Redirection and Piping history

Show command history

history > cmd.txt	# Save history to a file
echo 'New Entry' >> cmd.txt	# Append a new line to a
file head -4 file.txt tail -2	# Extract specific lines
man ls > lsmanpage	# Save manual page output to a file
cat -n lsmanpage	# Display file contents with line numbers

6. Text Transformation

<code>tr a-z A-Z < file.txt</code>	<code># Convert lowercase to uppercase.</code>
<code>tr ' ' '\n' < file.txt</code>	<code># Replace space with newlines.</code>
<code>tr -d 'a' < file.txt</code>	<code># Delete all occurrences of 'a'.</code>
<code>tr -s " < file.txt</code>	<code># Squeeze multiple spaces into one.</code>

(You can also use pipeline instead of “<”)

7. Word/Line/Character Counting

<code>wc file.txt</code>	<code># Count lines, words, and characters.</code>
<code>wc -l file.txt</code>	<code># Count lines only.</code>
<code>wc -w file.txt</code>	<code># Count words only.</code>
<code>wc -c file.txt</code>	<code># Count characters (bytes).</code>

8. Pipeline Examples

`# View sorted unique words in a file`
`cat file.txt | tr ' ' '\n' | sort | uniq`

`# Count how many times "Linux" appears in a file`
`grep 'Linux' file.txt | wc -l`

`# Extract the first 10 lines of history and count words`
`history | head -10 | wc -w`

`# Find all.txt files and count them`
`find . -type f -name "*.txt" | wc -l`

`# Convert file contents to uppercase and display first 5 lines`
`cat file.txt | tr a-z A-Z | head -5`

..... and many others. Practise with pipelines more.